MILITARY SPECIFICATION

CHARGE, PROPELLING, M204, PAPER PREFORM PROCESS PARTS AND CONTAINER ASSEMBLY

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the parts and container assembly for the propelling charge designated as Charge, Propelling, M204 manufactured by the paper preform process and used on the Lightweight Company Mortar System (LWCMS).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-810 - Environmental Test Methods

FSC: 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9277371 - Composition
9345311 - Container Assembly
9293287 - Packing and Marking Container, Ammunition, Fiber, PA 78 for 60MM Cartridge, HE, M720 with Fuze M734
9293439 - Packaging and Marking for Container Ammunition, Metal, PA 70 for 60MM Cartridge, HE, M720 with Fuze M734
9293440 - Box, Wirebound, Packing, Ammunition for 60MM Cartridge, HE, M720 with Fuze M734, Packing and Marking
9313703 - Packing and Marking for Box, Fiberboard, Outer for Charge Propelling, M204 or Container Assembly
9313727 - Packing for Pallet for Box, Fiberboard, Outer for Charge Propelling M204 or Container Assembly

(Copies of specifications, standards, drawing and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The materials including packing shall be in accordance with the applicable drawings and specifications.

3.2 Assemblies. Container assemblies shall comply with all the requirements specified on drawing 9345311 and on all related drawings and with all the requirements specified in the applicable specifications.

3.3 Stability heat test (134.5°C). The nitrocellulose in the container assemblies and closure shall not completely change the color of the methyl violet paper to salmon pink color in less than thirty-five minutes when tested in accordance with 4.5.2.

3.4 Back pressure test. The container assemblies shall satisfactorily pass the back pressure test specified in 4.5.4.

3.5 Residue test. The container assemblies when loaded and fired ballistically shall not leave residue which will cause cartridge hang-up in the mortar tube, shall not produce a secondary flame after 2 seconds (sec.) minimum (min.) from the primary flame of the cartridge launch and when fired at zone 4, shall not produce flaming debris on the ground . The test shall be performed as specified in 4.5.5.

3.6 Percent by weight of resin and additives, kraft and polyester staple fiber. The percent by weight of resin and additives, kraft and polyester, staple fiber shall not be less than 12.95 and not more than 21.55 when tested in accordance with 4.5.1.3.5.

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3.7 Light permeability. The top and bottom halves shall not exhibit cracks, splits, cuts, breaks, thin spots, holes or other inhomogeneities when tested as specified in 4.5.9. The Government shall establish visual standards prior to production with the assistance of the contractor.

3.8 Compression strength test. The container assemblies shall exhibit a compression strength of not less than 0.9 pounds and not more than 4.8 pounds when tested as specified in paragraph 4.5.6.

3.9 Sequential rough handling test. The container assemblies shall be able to withstand the sequential rough handling procedure of 4.5.7 and subsequently be safe to handle and fire.

3.10 Homogeneity of material. The material used to manufacture the parts and assemblies shall be uniform in structure or composition throughout. The presence of spots of 1/16 inch or larger diameter or any spot which completely permeates through the entire thickness of the part or assembly shall indicate non-homogeneity. The presence of creases or thin membrane-like areas shall also indicate non-homogeneity. The contractor and the Government shall establish visual standards prior to production.

3.11 Water repellancy. The container assemblies shall satisfactorily pass the water repellency test specified in 4.5.8.

3.12 Process description. A process control document, including a complete process description in accordance with DI-P-1604 (Modified) is required. Government approval of the process control document is required prior to initiation of production.

3.13 First article inspection. This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.14 Workmanship

3.14.1 Container parts and assemblies. The container parts and assemblies shall be free of dirt, grease or other foreign material.

3.14.2 Packaging. The packaging shall be free from foreign material. The boxes shall be assembled without undue binding and no flap shall project beyond the edge of the box when closed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawings	Quantity
Container, Top	9345313	15 per cavity
Container, Bottom	9345312	15 per cavity
Closure	9312707	200
Container Assembly	9345311	528
Container Assembly (less silicone)	9345311	35

The submission of 1st Article shall include 25 preforms (the cut-out from the paper roll stock that will be utilized in the die-press set-ups to form the increment tops and bottoms) as used in making the top and bottom halves submitted for 1st Article. Each preform shall be measured for thickness to 1/1000 inches in three places, for area to 1/100 square inches and for weight to 1/10 grams and the data submitted with the 1st Article identifiable to the particular preform.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I.

4.3.3 Rejection. See MIL-A-48078.

4.3.4 Initial qualification of die-press set-ups. Initial die-press set-up qualification shall be accomplished by conventional die maker techniques to check the concentricity of Male-Female dies. The following procedure shall be employed:

a. Strips of eutectic solder 0.032 inches in thickness are to be placed in the cavity of each die in three places. The dies are then to be closed on the strips. Subsequently the strips are to be removed from the dies and measured with a ball anvil and ball spindle micrometer. Inner diameter, outer diameter, and base thickness measurements are to be obtained and shall be in accordance with the applicable drawing.

b. After the die-press setups have been qualified with solder, fifteen each parts shall be processed through each die and measured for inner and

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outer diameters, and base thickness. All dimensions shall be in accordance with the applicable drawing.

c. All measurements obtained during die-press qualification shall be maintained on file and shall be available for government review. The die-press set-ups shall not be approved for use in production if any part, assembly or solder set-up fails to comply with above requirements.

d. All die-press set-ups shall operate against the established stop and meet an established dwell time.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63522 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Propelling Charge Parts Assembly				
	Container, Top (dwg. 9345313)				
	Examination for defects	15/Cav		3.2	4.4.2.2
	Container, Bottom (dwg. 9345312)				
	Examination for defects	15/Cav		3.2	4.4.2.3
	Closure (dwg. 9312707)				
	Examination for defects	15		3.2	4.4.2.4
	Stability Heat Test (134.5°C)	100		3.3	4.5.2
	Composition	100		3.2	4.5.1
	Container Assembly (dwg. 9345311)				
	Examination for defects	25		3.2	4.4.2.5
	Weight of silicone deposit	25		3.2	4.5.3
	Weight	350 *		3.2	Gage
	Back pressure test	350 *		3.4	4.5.4
	Compression strength test	350 *		3.8	4.5.6
	Homogeneity of material	350 *		3.10	Visual
	Water repellency	350 *		3.11	4.5.8
	Residue Test (Ballistic)	350 *		3.5	4.5.5
	Stability heat test (134.5°C)	5		3.3	4.5.2
	Composition	5		3.2	4.5.1
	Sequential Rough Handling Test	128		3.9	4.5.7

NOTES: * The same 350 samples may be used.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

a. Each lot of container tops and bottoms shall contain parts produced from:

(1) Nitrocellulose from not more than one lot from one manufacturer.
(2) Diphenylamine from not more than one lot from one manufacturer.
(3) Acrylic fibers from not more than one lot from one manufacturer.
(4) Polyester fibers from not more than one lot from one manufacturer.
(5) Kraft fibers from not more than one lot from one manufacturer.
(6) Resin from one manufacturer.
(7) Resin additives from one manufacturer.

b. Each lot of container assemblies shall contain parts produced from:

(1) Container tops and container bottoms from not more than one manufacturer and one lot.
(2) Silicone from one manufacturer.
(3) Hexane from one manufacturer.
(4) Acetone from one manufacturer.
(5) MR-23 from one batch from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Conditioning shall be at 23 + 2°C and 50 + 5% R.H. for a minimum of 4 hours. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2.1 Mold examination and process requirements.

4.4.2.1.1 Qualification for new or rework die-press set-ups. Qualification of new or reworked die-press set-ups shall be in accordance with 4.3.4. After initial qualification, the procedure shall be accomplished after any set-up change and at intervals of at least once per week per die.

4.4.2.1.2 Process requirements.

a. All assemblies shall be air-dried a minimum of two hours immediately after the assembly operation prior to further processing or inspection.

b. Light permeability inspection will be performed prior to assembly of container halves.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63522 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9345313
4.4.2.2	Container, Top				NEXT HIGHER ASSEMBLY 9345311
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Light permeability		100%	3.7	Visual (4.5.9)
Major					
101	Thickness of inside wall (average of 3 places)(a)		0.40%	3.2	Gage
102	Thickness of outside wall (average of 3 places)(a)		0.40%	3.2	Gage
103	Thickness of top (average 3 places)(a)		0.40%	3.2	Gage
Minor					
201	Height of flat		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.14.1	Visual

NOTES: (a) In addition, any top with an individual reading greater than .003 inches outside of drawing tolerance shall be considered defective. See paragraph 6.6 for alternate inspection.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63522 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Container, Bottom		9345313
			NEXT HIGHER ASSEMBLY 9345311

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Light permeability		100%	3.7	Visual (4.5.9)
Major					
101	Thickness of inside wall (average of 3 places)(a)		0.40%	3.2	Gage
102	Thickness of outside wall (average of 3 places)(a)		0.40%	3.2	Gage
103	Thickness of bottom (average of 3 places)(a)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.14.1	Visual

NOTES: (a) In addition, any bottom with an individual reading greater than .003 inches outside of drawing tolerance shall be considered defective. See paragraph 6.6 for alternate inspection.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63522 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Closure	1 OF 1	9312707
			NEXT HIGHER ASSEMBLY 9345311

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Length		0.65%	3.2	Gage
202	Width		0.65%	3.2	Gage
203	Thickness		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.14.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63522 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9345311
4.4.2.5	Container Assembly				NEXT HIGHER ASSEMBLY 9312698
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight		100%	3.2	Gage
102	Back pressure test		100%	3.4	4.5.4
103	Weight of silicone deposit	10	(a)	3.2	4.5.3
104	Composition			3.2	4.4.3.1/4.5.1
105	Stability heat test			3.3	4.4.3.2/4.5.2
106	Outside diameter, max. (d)		0.40%	3.2	Gage
107	Inside diameter, min. (d)		0.40%	3.2	Gage
108	Distance across open end, max.		0.40%	3.2	Gage
109	Height		0.40%	3.2	Gage
110	Compression strength test	(b)	0.40%	3.8	4.5.6
111	Residue test (ballistic)(c)	150		3.5	4.5.5
112	Homogeneity		100%	3.10	Visual
113	Water repellency test		0.15%	3.11	4.5.8
Minor					
201	Diameter of loading hole (prior to leak test)		0.65%	3.2	Gage
202	True position of closure seat		0.65%	3.2	Gage
203	Width of closure tab		0.65%	3.2	Gage
204	True position of loading hole		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.14.1	Visual

NOTES: (a) One or more assemblies that fails to meet the weight requirement shall be cause for lot rejection. (b) The sample size and acceptance criteria shall be in accordance with MIL-STD-414, Inspection Level II, Section B, Part II with an AQL of 0.40%. (c) After the first successful production lot, discontinue residue testing. (d) Additionally, outside diameter, min, and inside diameter, max. shall be inspected for one lot out of every five lots and during 1st article in accordance with MIL-STD-105, Level II at an AQL of 0.40%.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63522 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Packing and Marking for Box, Fiberboard, Outer for Charge, Propelling, M204 or Container Assembly	1 OF 1	9313703
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Containers incorrectly packed		0.40%	5.1	Visual
102	Marking improper		0.40%	5.1	Visual
103	Improper seal		0.40%	5.1	Visual
104	Orientation stripe on corner of outside box improperly located (with respect to loading holes of charges)*		0.40%	5.1	Visual
105	Folded bag in inner box missing (prior to sealing inner box)		0.40%	5.1	Visual
106	Barrier bag damaged (prior to inserting bagged inner box into outer box)		0.40%	5.1	Visual
107	Barrier bag closed improperly (prior to sealing outer box)		0.40%	5.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.17.2	Visual

NOTES:

* Applicable to automated charge loading line only.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63522 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Packing for Pallet				9313727
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Orientation of boxes improper		0.40%	5.2	Visual
102	Strapping loose		0.40%	5.2	Visual
103	Boxes damaged		0.40%	5.2	Visual
Minor	None defined				

NOTES:

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4.4.3 Testing

4.4.3.1 Composition (see dwg. 9277371) major defect. Five container assemblies and one hundred closures shall be selected from each lot and tested in accordance with 4.5.1. The container assemblies shall be selected just prior to the application of silicone. The lot shall be rejected if the sample fails to comply with the requirements.

4.4.3.2 Stability heat test (134.5°C) (see 3.3) major defect. Five container assemblies and one hundred closures shall be selected from each lot and tested in accordance with 4.5.2. The container assemblies shall be selected just prior to the application of silicone. The lot shall be rejected if the sample fails to comply with the requirements.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the "Quality Conformance Inspection" tables. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures

4.5.1 Composition

4.5.1.1 Preparation of samples. Each sample shall be cut into approximately 1/4 inch squares. From each sample, approximately 0.5 gram shall be selected for testing and weighed to the nearest 0.2 milligram (Weight A) in an extraction thimble. The same thimble shall be utilized during the extractions in 4.5.1.3.1 and 4.5.1.3.3. Test is destructive and samples should be removed from the lot.

4.5.1.2 Total volatiles. The samples from 4.5.1.1 shall be dried for 3 hours at 100 ± 2°C, cooled in a desicator (1/2 hr. min.), and weighed to the nearest 0.2 mg. (weight W). Calculate the percent total volatiles as follows:

$$\text{Percent (\%) Total Volatiles} = \frac{A - W}{A} \times 100$$

4.5.1.3 Analytical procedure

4.5.1.3.1 Solvent extraction. Using a tared, extracoarse, sintered glass thimble, extract each sample from 4.5.1.2, with methylene chloride, for a minimum of 16 hours in a Roweg extractor (MIL-STD-286 Method 104.1.3). Evaporate the extract to dryness, using a current of clean, dry air. This extract is to be used for diphenylamine determination in 4.5.1.3.2. Dry the residues (non-extractable material) in the extraction thimbles by aspirating the thimble and contents until the odor of methylene chloride is not discernable. Dry the thimble plus residue for 1/2 hour minimum at 100 ± 2°C. Cool in a desicator (1/2 hr. min.). Determine the dry weight of this residue (Weight C) to

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the nearest 0.2 mg. as the difference between this weight and the weight of the empty thimble. This residue is to be used for the nitrocellulose determination in 4.5.1.3.3.

4.5.1.3.2 Diphenylamine (DPA). To the dried extract from 4.5.1.3.1 add 50 milliliters (ml.) of glacial acetic acid. If necessary, warm slightly to dissolve the extract, then cool to room temperature. Complete the determination of diphenylamine in the solution according to Method 201.1.4 (starting with paragraph 5.5) of MIL-STD-286 except that 25 ml. of methylene chloride is to be added after addition of the potassium iodide solution. When the end point appears to have been reached, (absence of blue color) the flask is stoppered and shaken. If no color appears, the end point has been reached; otherwise, continue titration until the blue color is discharged and shaking produces no further color. Calculate the percent DPA as shown in MIL-STD-286, Method 201.1.4, paragraph 5.13.

4.5.1.3.3 Nitrocellulose. To provide good heat conduction to the solvent, place a sufficient quantity of glycerin on the flask bottom and on the surface of the heat source to produce a continuous film. Add 10 ml. of ethyl alcohol to the thimble and residue from 4.5.1.3.1 and extract with acetone using the method in 4.5.1.3.1. After extraction dry the residue remaining at 100 + 2°C for one hour. Cool in a desiccator (1/2 hr. min), weight to the nearest 0.2 mg. (Weight D) and save for the acrylic determination in 4.5.1.3.4. Evaporate the acetone extract to dryness and add 25 ml. of n-butyl acetate and 45 ml of glacial acetic acid. Replace the air with a stream of inert gas (CO_2 or N_2). Maintain the inert atmosphere throughout this test. Reflux the solution for five minutes. Add to the flask, via the condenser, 25 ml of 0.7N ferrous ammonium sulfate (prepared per MIL-STD-286, Method 604.1.1) and 10 ml. concentrated hydrochloric acid. Reflux for thirty minutes more to obtain a yellow liquid in which the specimen is completely disintegrated. With the inert gas still running, cool in a water bath to room temperature. Add 5 ml. of 20% ammonium thiocyanate solution (MIL-STD-286, Method 601.1.1). Titrate with a 0.2 titanous chloride solution (MIL-STD-286, Method 707.1). Perform a blank titration on the reagents. Calculate percent NC as follows:

$$\text{Percent (\%) NC} = \frac{(46.69)\ (N)\ (V-B)}{(EXW)} \times 100$$

Where:

V = Volume titanous chloride used for sample, ml.
B = Volume titanous chloride used for blank, ml.
W = Weight of dried sample, gm.
N = Normality of titanous chloride
E = Percent nitrogen in the nitrocellulose

4.5.1.3.4 Acrylic

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Reagents

SULFURIC ACID - 93 - 98% H_2SO4, N_2 Free

MEMCURIC OXIDE (H_gO) - Reagent Grade, N_2 Free

POTASSIUM SULFATE OR ANHYDROUS SODIUM SULFATE - Reagent Grade, N_2 Free

THIOSULFATE SOLUTION - 160 g $Na_2S_2O_3$ $5H_2O$ dissolved in one liter of H_2O

SODIUM HYDROXIDE - 400 g solid NaOH dissolved in H_2O and diluted to one liter

METHYL RED INDICATOR - 1 g methyl red dissolved in 200 ml ethyl alcohol

METHYLENE BLUE INDICATOR - 1 g methylene blue dissolved in 200 ml ethyl alcohol

HYDROCHLORIC OR SULFURIC ACID STANDARD SOLUTION - 0.1N

BORIC ACID SOLUTION - 4 g H_3BO_4 dissolved in 100 ml H_2O

HENGAR GRANULES

Apparatus

DIGESTION - Use Kjeldahl flasks of hard, moderately thick well-annealed glass with total capacity of 500 to 800 ml. Conduct digestion over a safety-approved heating device adjusted to bring 250 ml. H_2O at 25°C to a rolling boil in five minutes.

DISTILLATION - Use a Kjeldhal flask of 500 to 800 ml. capacity fitted with a rubber stopper. The low end of an efficient scrubber bulb or trap is passed through the rubber stopper to prevent mechanical carryover of NaOH during distillation. Use rubber tubing to connect the upper end of the bulb to the conderser tube. Trap the condenser outlet to insure complete absorption of distilled NH_3 into boric acid in the receiver.

Analysis

Quantitatively transfer the residue from 4.5.1.3.3 to a Kjeldahl flask, add 3-4 Hengar granules, 0-7g. of HgO, 15.0g. of K_2SO_4 or Na_2SO_4, and 25 ml H_2SO_4 to the flask. Place the flask in an inclined position, and heat gently until frothing ceases. Boil the contents of the flask briskly until the solution clears. Continue boiling for 1/2 hours minimum. Allow the solution to cool to room temperature. Add 200 ml. H_2O. Cool the solution to below 25°C. Add 25 ml. thiosulfate solution and mix to precipitate the mercury. Tilt the flask and add a layer of 100 ml. NaOH solution without agitation. Immediately connect the flask to the distilling bulb on the condenser. With the tip of the condenser immersed in the boric acid in the

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receiver, rotate the flask to mix the contents thoroughly. Heat the flask until all of the NH_3 has distilled, (approximately 250 ml.) Titrate the boric acid solution with the standard acid using a mixed indicator containing 4 drops of methyl red, and 2 drops of methylene blue solution. Correct for the reagent blank. Calculate the percent acrylic as follows:

$$\text{Percent (\%) Acrylic} = \frac{(6.05)\ (V-B)\ (N)}{W} \times 100$$

Where:

V = Volume of standard acid for sample, ml.
B = Volume of standard acid for blank, ml.
N = Normality of standard acid
W = Dried sample weight, g.

4.5.1.3.5 Resin and additives, kraft and polyester staple fiber. Calculate the percent resin and additives, kraft and polyester staple fiber as follows:

% Resin + % additives + % kraft + % polyester staple fiber = 100 - (% NC + % Acrylic + % DPA)

4.5.1.3.6 Methylene Chloride Extractables. Calculate the total methylene chloride extractables $\frac{(W-C)}{W} \times 100$ and the

total acetone extractables $\frac{(C-D)}{W} \times 100$

4.5.2 Stability heat test (134.5°C). Samples shall be tested in accordance with MIL-STD-286, Method 404.1.2, Heat Tests at 134.5°C. Test is considered destructive and samples should be removed from the lot.

4.5.3 Weight of silicone deposit. Each sample container assembly less closure shall be weighed individually before and after silicone application. The results of these two weights shall determine the deposit percent by weight of silicone for each sample. Failure of any sample to comply with the requirement is cause for lot rejection.

4.5.4 Back pressure test. The container assemblies shall be subjected to an internal pressure of 5.0 psig pressure, minimum. Observation shall be made for back pressure as indicated on the air pressure source and container assembly as described below. Acceptable containers shall be indicated by a back pressure reading of greater than 5.6 psig. Indication of a leaking container shall be a back pressure reading of 5.6 psig, or less; these will be segregated and subjected to a visual inspection for open seams or leaks. Those with no visible leaks will be accepted if, upon retest a back pressure reading of 5.3 psig or greater is obtained.

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4.5.4.1 Back pressure test. The back pressure test pressure source shall be clean dry air or nitrogen at 20 psig. The pressureline set-up shall consist of an adjustable pressure restrictor (throttleable to 5 psig), a pressure gage, connected by a "T" and a test nozzle located at the end of the line. A gas cylinder and two-stage regulator equipped with a needle valve may be used. The pressure gage shall have a scale indication of zero to 10 psi with 0.1 psi graduation or smaller. Readings shall be taken to the nearest 0.1 psi. The nozzle shall consist of a 1/4 inch diameter steel tube with a 5/32 inch I.D. opening, a length of approximately 1 1/2 inch, with a stop and silicone "O" ring about 1/4 inch from the test end. Care shall be exercised when inserting the nozzle into the opening of the contained assembly to avoid damage. "The regulator is adjusted to deliver a 5.0 psig reading on the pressure gage". Any assembly that fails to meet the requiremnts of 4.5.4 shall be classified defective and removed from the lot. Data results of this test shall be submitted in accordance with DI-T-1906.

4.5.5 Residue test. The test shall be performed at a Government Proving Ground. Empty containers will be loaded with M10 propellant and sealed in accordance with dwg. 9312698. Loaded containers shall then be assembled to inert-filled cartridges in accordance with the appropriate zones as listed in Tables 2 or 3. Test rounds shall be conditioned for a minimum of eight hours at 70°F plus or minus 5°F and fired within two minutes after removal from conditioning environment. Rounds shall be fired from a M224 mortar (standard or ballistic tube) at a firing elevation of 800 mils, at a uniform rate of twelve to fifteen rounds per minute. The tube shall be dry swabbed and the elevation checked between the firing of each group.

Observation shall be made for hangups, secondary flame, and flaming debris. The lot shall be accepted if no hangups occur, no rounds exhibit secondary flame at any zone, and no flaming debris reach the ground at zone 4 only.

If a hangup occurs, the test round shall be removed from the tube and an examination shall be conducted of the tube and test round to determine if the hangup was caused by residue. If the hangup was not caused by residue, the test shall be continued. If the hangup was caused by residue, firing of test rounds shall be discontinued. The tube shall be dry swabbed and a group of reference rounds shall be immediately fired.

The reference round group shall consist of 25 rounds of the same zone, configuration, and conditioning as the test round except that loaded containers from an accepted lot shall be used. The reference group shall be fired after dry swabbing from the same tube used for the test groups. If no hangup occurs in the reference group, the test lot shall be referred.

The test lot shall also be rejected if one or more rounds exhibit secondary flame at any zone or flaming debris reach the ground during zone 4 firing. All other flaming debris results shall be reported for information only. A formal firing report is required.

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4.5.5.1 First article. One hundred and seventy five test rounds shall be ballistically tested. These rounds shall consist of seven groups. Each group shall be zone fired in the sequence indicated in Table 2.

TABLE 2

Group	M204 Zone	Test Rounds
1	1	25
2	2	25
3	3	25
4	4	25
5	1	25
6	2	25
7	1	25

4.5.5.2 Production. Seventy-five test rounds shall be ballistically tested. These rounds shall consist of three groups. Each group shall be zone fired in the sequence indicated in Table 3.

TABLE 3

Group	M204 Zone	Test Rounds
1	1	25
2	4	25
3	1	25

4.5.6 Compression strength test. The empty container assembly shall be inserted between the plates of a spring compression tester in such a manner that a straight line drawn through the ends of the container would be perpendicular to the plane of the plates. With a .606 inch precision block held in the opening between the ends of the container assembly, apply a compression load to the container assembly until the container ends make contact with the precision block. The reading at the initial point of contact with the precision block shall be recorded. Results shall be recorded and evaluated to determine compliance with the requirements. The application of the compression load shall be accomplished within 15 seconds, maximum. Data results from this test will be submitted in accordance with DI-T-1906.

4.5.7 Sequential rough handling. (First Article Only) Container assemblies shall be loaded and assembled to thirty-two inert M720 cartridges, the remaining cartridges required to complete the packing may consist of inert M720 cartridges without container assemblies. Cartridges shall be packed in accordance with dwgs. 9293287, 9293439 and 9293440 except that four cartridges with loaded container assemblies shall be packed in each container as per dwg. 9293439 with one cartridge being placed in the corner positions of the container. Sixteen rounds (two boxes) shall be conditioned to -50°F, min.

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and sixteen rounds (two boxes) shall be conditioned to +145°F, min. for a minimum of 24 hours. The cartridges shall be subjected to electrodynamic transportation-vibration in accordance with Method 514.2, Procedure X of MIL-STD-810 for simulated truck and simulated two-wheel trailer transportation. The test shall be conducted at a goverment-approved facility. Test cartridges shall remain boxed (packaged) throughout the test. Cartridges shall be tested on two axes, i.e., 1 box vertically per temperature condition and 1 box longitudinally per temperature condition. The boxes shall be unloaded after testing and the container assemblies shall be visually inspected. Evidence of broken container assemblies or any propellant spillage shall cause the container assembly lot to be rejected.

4.5.8 Water repellency test. The container assemblies shall be tested for the presence of silicone by brief immersion in water (1-2 seconds) and inspection for "beading" or non-wetting. The container is held between the thumb and finger in such a way as to cover the fill hole and permit complete immersion of the container; the assembly shall then be withdrawn and examined. A container shall be rejected if it shows a lack of silicone or incomplete coverage as evidenced by wetting or lack of beading. Alternative: Where desired, water repellency may be determined by weight pickup after immersion and shaking to remove excess water. A weight pickup of greater than 3% shall cause the container to be rejected.

4.5.9 Light permeability. Tops and bottoms shall be light table inspected. Any item which fails to conform with the light permeability requirement shall be classed defective and removed from the lot.

5. PREPARATION FOR DELIVERY

5.1 Unit packing. Packing shall be as specified on dwg. 9313703.

5.2 Exterior packing. Packing shall be as specified on dwg. 9313727.

5.3 Marking. Marking shall be as specified on dwg. 9313703 and dwg. 9313727.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 60MM Lightweight Company Mortar System (LWCMS).

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approval. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, NJ 07801.

6.4 Drawings. Drawings listed in section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may

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also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Additives. Additives shall have prior approval by the Contracting Officer. Mold release agents and processing aids are considered additives.

6.6 Alternate thickness inspection. At the discretion of the Government contracting officer, the contractor may elect to utilize the following inspections in lieu of the thickness inspectons of 4.4.2.2 and 4.4.2.3. The preform shall be inspected for area by template inspection to MIL-STD-105 and an AQL of 0.40%. The requirement shall be that the preform edges shall be within two areas outlined on a template that represent areas of plus or minus 1% of the average area of the 25 preforms submitted during 1st Article. In addition, the same sample shall be measured for thickness in three places and shall each have an average between ± .003 inches of the average of the 25 preforms submitted during 1st Article. No individual reading shall be outside of ± .005 inches of the average of the 25 preforms submitted during 1st Article. In addition, each preform from the sample shall be weighed and shall be within ± .2 grams of the average weight of the 25 preforms submitted during 1st Article. The above alternative does not relieve the contractor from die-press qualification of 4.3.4 and 4.4.2.1.1.

6.7 Hangup. For this specification, the meaning of a "hangup" is a round which is dropped into the mortar tube but does not strike the firing pin with enough force to function the primer.

6.8 Submission of formal proving ground firing report. Residue test data shall be sent to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-A, Dover, NJ 07801.

6.9 Proving ground test summary

Test	Sample Size	Requirement
Residue Test (First Article)	350 Containers (175 test rounds, 25 reference rounds	Absence of hangup, secondary flame, & flaming debris, zone 4
Residue Test (Production)	150 Containers (75 test rounds)	Absence of Hangup, secondary flame, & flaming debris, zone 4

6.10 Data requirements. Deliverable data required by this specification and not referenced elsewhere are cited below. Data shall be furnished to DRDAR-QAT-A, ARRADCOM, Dover, NJ 07801.

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Paragraph	Data Requirement	Applicable DID
3.12	Process Description	DI-P-1604
4.5.4.1	Test & Demonstration Reports	DI-T-1906
4.5.6	Test & Demonstration Reports	DI-T-1906

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